Table 1: Exploratory linear–neural simulation extension: 10 paired repetitions per setting and design. Neural minus matched linear uPU/nnPU is shown separately for risk-based and proxy-AUROC hyperparameter selection. Bars retain the saved pointwise paired t intervals (9 degrees of freedom), not Gaussian large-sample intervals. No BA headline is drawn; these capacity comparisons do not isolate boosting.

One-sample - Reference					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	-0.32	[-0.46, -0.19]	-1.16	[-1.37, -0.94]	10
nnPU / risk	-0.34	[-0.51, -0.18]	-1.06	[-1.29, -0.84]	10
uPU / proxy-AUROC	-0.32	[-0.46, -0.19]	-1.16	[-1.37, -0.94]	10
uPU / risk	-0.34	[-0.51, -0.18]	-1.06	[-1.29, -0.84]	10

One-sample - Labeling probability $c = 0.05$					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	-2.48	[-4.28, -0.68]	-3.89	[-5.93, -1.85]	10
nnPU / risk	-2.04	[-2.75, -1.33]	-3.24	[-4.24, -2.25]	10
uPU / proxy-AUROC	-2.53	[-4.42, -0.64]	-3.91	[-5.98, -1.84]	10
uPU / risk	-2.04	[-2.75, -1.33]	-3.24	[-4.24, -2.25]	10

One-sample - High dimension ($p = 500$)					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	-0.28	[-0.44, -0.13]	-0.15	[-0.32, +0.03]	10
nnPU / risk	+6.26	[+5.19, +7.33]	+4.45	[+3.80, +5.10]	10
uPU / proxy-AUROC	-1.28	[-1.68, -0.88]	-1.40	[-1.72, -1.08]	10
uPU / risk	+6.26	[+5.19, +7.33]	+4.45	[+3.80, +5.10]	10

One-sample - Curved boundary					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+0.93	[+0.69, +1.17]	+1.41	[+0.81, +2.00]	10
nnPU / risk	+0.92	[+0.75, +1.09]	+1.22	[+0.74, +1.70]	10
uPU / proxy-AUROC	+0.93	[+0.69, +1.17]	+1.41	[+0.81, +2.00]	10
uPU / risk	+0.92	[+0.75, +1.09]	+1.22	[+0.74, +1.70]	10

One-sample - Feature interaction					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+8.32	[+2.78, +13.86]	+8.69	[+4.85, +12.52]	10
nnPU / risk	+5.88	[+5.33, +6.43]	+7.08	[+6.40, +7.76]	10
uPU / proxy-AUROC	+8.32	[+2.78, +13.86]	+8.69	[+4.85, +12.52]	10
uPU / risk	+5.88	[+5.33, +6.43]	+7.08	[+6.40, +7.76]	10

One-sample - Unequal class covariance

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+0.13	[-0.37, +0.63]	-0.19	[-0.86, +0.49]	10
nnPU / risk	+0.27	[+0.00, +0.54]	-0.01	[-0.35, +0.32]	10
uPU / proxy-AUROC	+0.13	[-0.37, +0.63]	-0.19	[-0.86, +0.49]	10
uPU / risk	+0.27	[+0.00, +0.54]	-0.01	[-0.35, +0.32]	10

Two-sample - Reference

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	-0.33	[-0.49, -0.17]	-1.16	[-1.54, -0.78]	10
nnPU / risk	-0.35	[-0.56, -0.14]	-1.04	[-1.38, -0.70]	10
uPU / proxy-AUROC	-0.33	[-0.49, -0.17]	-1.16	[-1.54, -0.78]	10
uPU / risk	-0.35	[-0.56, -0.14]	-1.04	[-1.38, -0.70]	10

Two-sample - Labeled positives $n_P = 50$

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	-1.52	[-2.09, -0.95]	-2.47	[-3.58, -1.36]	10
nnPU / risk	-1.68	[-2.21, -1.14]	-2.76	[-3.54, -1.97]	10
uPU / proxy-AUROC	-1.52	[-2.09, -0.95]	-2.47	[-3.58, -1.36]	10
uPU / risk	-1.68	[-2.21, -1.14]	-2.76	[-3.54, -1.97]	10

Two-sample - High dimension ($p = 500$)

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+2.36	[-1.05, +5.77]	+1.80	[-0.57, +4.17]	10
nnPU / risk	+5.72	[+4.28, +7.17]	+3.72	[+2.78, +4.67]	10
uPU / proxy-AUROC	+0.81	[-2.74, +4.36]	+0.16	[-2.30, +2.61]	10
uPU / risk	+5.72	[+4.28, +7.17]	+3.72	[+2.78, +4.67]	10

Two-sample - Curved boundary

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+0.96	[+0.60, +1.32]	+1.00	[+0.49, +1.51]	10
nnPU / risk	+1.01	[+0.72, +1.30]	+1.17	[+0.75, +1.58]	10
uPU / proxy-AUROC	+0.96	[+0.60, +1.32]	+1.00	[+0.49, +1.51]	10
uPU / risk	+1.01	[+0.72, +1.30]	+1.17	[+0.75, +1.58]	10

Two-sample - Feature interaction

Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+5.87	[+5.16, +6.57]	+7.27	[+5.71, +8.83]	10
nnPU / risk	+5.80	[+5.16, +6.43]	+7.02	[+5.62, +8.43]	10
uPU / proxy-AUROC	+5.87	[+5.16, +6.57]	+7.27	[+5.71, +8.83]	10
uPU / risk	+5.80	[+5.16, +6.43]	+7.02	[+5.62, +8.43]	10

Two-sample - Unequal class covariance					
Matched contrast	AUROC (pp)	AUROC interval	AP (pp)	AP interval	n
nnPU / proxy-AUROC	+0.11	[-0.27, +0.49]	-0.08	[-0.79, +0.63]	10
nnPU / risk	-0.04	[-0.34, +0.27]	-0.57	[-1.26, +0.12]	10
uPU / proxy-AUROC	+0.11	[-0.27, +0.49]	-0.08	[-0.79, +0.63]	10
uPU / risk	-0.04	[-0.34, +0.27]	-0.57	[-1.26, +0.12]	10

Note. Exploratory: 10 paired repetitions per setting. Neural minus linear with matched risk and selection criterion. Intervals are the existing paired t intervals (df=9); they are retained as saved. Proxy-AUROC denotes the PU validation tuning criterion, not test-data selection.